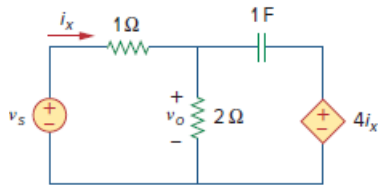


Problem

The initial energy in the circuit of Fig. 16.17 is zero at $t = 0$. Assume that $v_s = 360u(t)$ V. (a) Find $V_o(s)$ using the Thevenin theorem. (b) Apply the initial- and final-value theorems to find $v_o(0)$ and $v_o(\infty)$. (c) Obtain $v_o(t)$.

Figure 16.17:



Step-by-step solution

Step 1 of 12

(a)

There is no initial energy stored in the circuit, hence the initial capacitor voltage is zero at $t = 0$. Find the Thevenin's equivalent circuit. Remove the resistor across which the voltage is to be determined and represent the circuit in s domain as shown below:

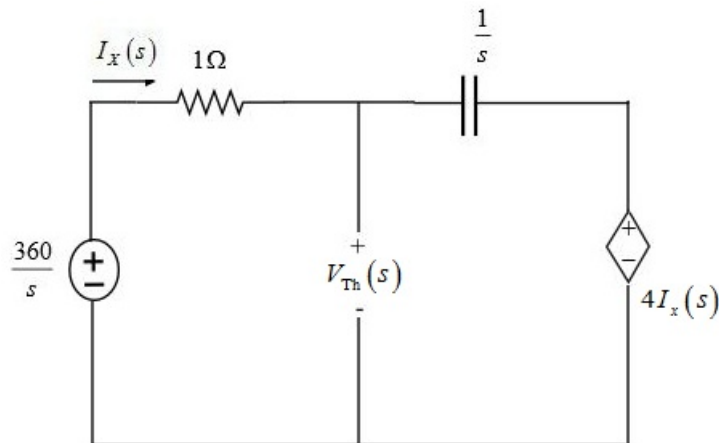


Figure 1

Step 2 of 12

Apply the Kirchhoff voltage law in the circuit shown in figure 1 to find current flowing through the circuit $I_x(s)$ as shown below

$$-\frac{360}{s} + I_x(s) + \left(\frac{1}{s}\right)I_x(s) + 4I_x(s) = 0 \quad \dots\dots (1)$$

Therefore,

$$I_x(s) = \frac{360}{5s+1} \quad \dots\dots (2)$$

Step 3 of 12

Find the Thevenin voltage in s-domain for the circuit shown in figure 1.

Redraw the circuit as shown in figure 2.

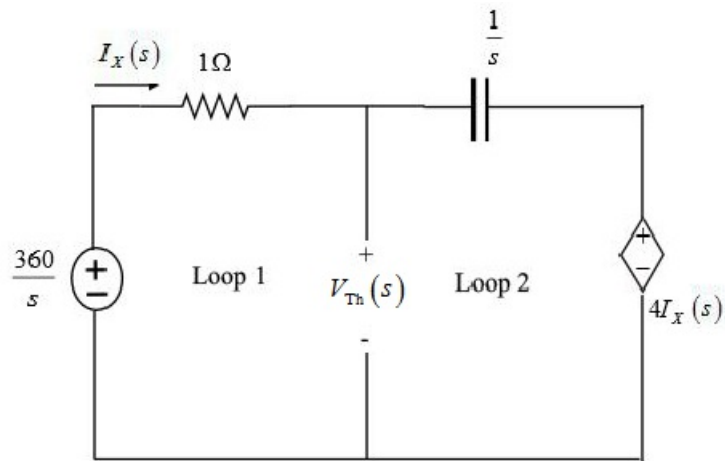


Figure 2

Step 4 of 12

Apply Kirchhoff voltage law in loop 1 of the circuit shown in figure 2 to find open circuit voltage

$V_{Th}(s)$.

$$-\frac{360}{s} + I_x(s) + V_{Th}(s) = 0$$

$$V_{Th}(s) = \frac{360}{s} - \frac{360s}{s(5s+1)}$$

$$V_{Th}(s) = \frac{360}{s} \left(\frac{4s+1}{5s+1} \right)$$

Step 5 of 12

Determine short circuit current I_{sc} . Replace the load resistance with short circuit as shown in figure 3.

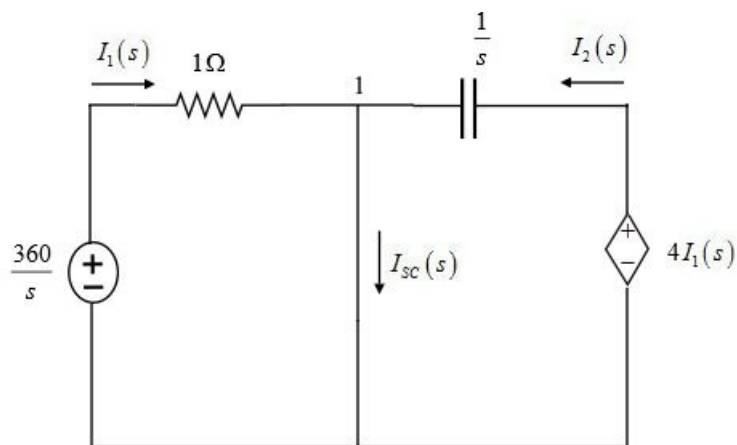


Figure 3

Step 6 of 12

Apply Kirchhoff current law at node 1 of the circuit represented in figure 3.

$$I_{sc}(s) = I_1(s) + I_2(s)$$

$$I_{sc}(s) = \frac{360}{s} + \frac{4I_1(s)}{\left(\frac{1}{s}\right)}$$

$$I_{sc}(s) = \frac{360}{s} + 4s\left(\frac{360}{s}\right)$$

$$I_{sc}(s) = \frac{360}{s} + 1440$$

Step 7 of 12

Determine the Thevenin's impedance $Z_{Th}(s)$.

$$Z_{Th}(s) = \frac{V_{Th}(s)}{I_{sc}(s)} \dots\dots (3)$$

Substitute the expressions of $V_{Th}(s)$ and $I_{sc}(s)$ in equation (3).

$$Z_{Th}(s) = \frac{\frac{360}{s} \left(\frac{4s+1}{5s+1} \right)}{\frac{360+1440s}{s}}$$

$$= \frac{1}{5s+1}$$

Step 8 of 12

Draw the Thevenin's equivalent circuit with the load resistance of 2Ω as shown in figure 4.

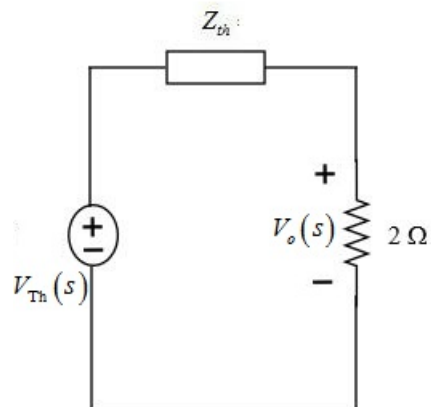


Figure 4

Step 9 of 12

Find the voltage across the resistance of 2Ω shown in figure 4.

Apply the voltage division rule to the circuit shown in figure 4.

$$\begin{aligned}V_o(s) &= V_{Th}(s) \left(\frac{2}{2 + Z_{Th}(s)} \right) \\&= \frac{360}{s} \left(\frac{4s+1}{5s+1} \right) \left(\frac{2}{\frac{1}{5s+1} + 2} \right) \\&= \left(\frac{720}{s} \right) \left(\frac{4s+1}{5s+1} \right) \left(\frac{5s+1}{10s+3} \right) \\&= \frac{288(s+0.25)}{s(s+0.3)}\end{aligned}$$

Therefore, voltage $V_o(s)$ appearing across 2Ω resistor is,

$$V_o(s) = \frac{288(s+0.25)}{s(s+0.3)}.$$

Step 10 of 12

(b)

Apply the initial value theorem to find initial voltage $v_o(0)$ as shown below

$$\begin{aligned}v_o(0) &= \lim_{s \rightarrow \infty} s V_o(s) \\&= \lim_{s \rightarrow \infty} s \frac{288(s+0.25)}{s(s+0.3)} \\&= \lim_{s \rightarrow \infty} \frac{288s \left(1 + \frac{0.25}{s} \right)}{s \left(1 + \frac{0.3}{s} \right)} \\&= 288 \text{ V}\end{aligned}$$

Therefore, the initial value $v_o(0)$ is $\boxed{288 \text{ V}}$.

Step 11 of 12

Apply the final value theorem to find final voltage $v_o(\infty)$.

$$\begin{aligned}v_o(\infty) &= \lim_{s \rightarrow 0} s V_o(s) \\&= \lim_{s \rightarrow 0} s \frac{288(s+0.25)}{s(s+0.3)} \\&= \frac{288(0.25)}{0.3} \\&= 240 \text{ V}\end{aligned}$$

Therefore, the final value $v_o(\infty)$ is $\boxed{240 \text{ V}}$.

Step 12 of 12

(c)

Resolve voltage $V_o(s)$ appearing across 2Ω resistor into partial fractions.

$$\frac{288(s+0.25)}{s(s+0.3)} = \frac{A}{s} + \frac{B}{s+0.3}$$

$$288(s+0.25) = A(s+0.3) + Bs$$

Equate s coefficients on both sides.

$$A + B = 288 \dots\dots (4)$$

Equate constants on both sides.

$$0.3A = 72 \dots\dots (5)$$

Solve equations (4) and (5) to find A and B and substitute them in partial fraction expansion as shown below

$$V_o(s) = \frac{240}{s} + \frac{48}{s+0.3} \dots\dots (6)$$

Take inverse Laplace transform of equation (6) to find $v_o(t)$.

Therefore,

$$v_o(t) = (240 + 48e^{-0.3t})u(t) \text{ V}$$

Therefore, the output voltage in time domain is $\boxed{(240 + 48e^{-0.3t})u(t) \text{ V}}$.